

Quantifying mobile robot localization safety for an EKF-based SLAM estimator: An integrity monitoring approach

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Abstract

The safety of SLAM-based estimation for mobile robots is a challenging research problem, particularly for life- or mission-critical exploratory applications. To address this problem, this work proposes a method to utilize integrity risk, a widely used performance metric in aviation, to quantify SLAM-based mobile robot's localization safety. More importantly, the approach accounts for sensor measurement faults, unknown deterministic errors that cannot be modeled via Gaussian white noise. The method is tailored for an EKF-based SLAM estimator, a chi-squared failure detector, and a local nearest neighbors data association criterion. The results show that data association errors can cause significant positioning performance degradation that can only be predicted using the proposed integrity risk metric. Furthermore, the study demonstrates that as the map's landmark density increases, mobile robot localization safety improves, except when the landmark map's density is too high to make the features indistinguishable, which leads to positioning safety degradation.

Keywords

Simultaneous localization and mapping, Kalman filtering, integrity monitoring, probability and statistical modeling, autonomous vehicles, robot safety

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1. Introduction

In the past decade, there have been significant developments in highly autonomous vehicles, as shown by the increasing number of safely driven miles on public roads. For instance, Waymo has conducted drive experiments totaling more than 40 million miles (The Waymo Team, 2024). Despite such a progression, such experiments are extremely insufficient to prove autonomous vehicle's safety, which requires driving billions of accident-free miles in all foreseeable conditions to provide credible proof of safety, all without altering the onboard algorithms (Kalra and Paddock, 2016). Simulation scenarios can partially address such a limitation, but they are unable to consider the complexities of real-world conditions (Wiggers, 2020). As a result, instead of experimental proof, rigorous analytical approaches may be necessary to ensure self-driving vehicle's overall system safety. Accordingly, this paper quantifies the safety of robot's localization subsystem. Specifically, it is the first to provide safety upper-bounds for mobile robot's localization using extended Kalman filter-based simultaneous localization and mapping (EKF-based SLAM) under the impact of data association failures modes.

This paper focuses on EKF-based SLAM because most, if not all, state-of-the-art localization and mapping techniques are currently too complicated to be utilized in rigorous safety analyses via the integrity risk. Instead, this paper approaches the problem from the perspective of the aviation due to its strict safety needs. Specifically, this industry relies on EKF-based techniques because of their suitability for integrity monitoring analysis (Langel et al., 2015; Tanil et al., 2018; Tanil, 2022). Thus, we contend that a critical focus on safety is needed in self driving, and so the community should first tackle EKF-SLAM before adapting

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these safety methods to more modern SLAM state estimators (Cadena et al., 2016).

The vast majority of publications in mobile robotic positioning techniques focus on enhancing SLAM performance (Dellaert and Kaess, 2017; Gallego and Scaramuzza, 2017; Dubé et al., 2018; Hsiao and Kaess, 2019; Jiwoong and Woojin, 2018); few papers have tackled the safety quantification of mobile robot's localization algorithms. Such papers typically used a sigma covariance bounds (Roumeliotis and Rekleitis, 2003; Bonnabel et al., 2016) to assess localization safety, which is proven to be a non-guaranteed safety metric for life-critical applications because they do not account for events in which measurements are faulted (Ahmad and Namerikawa, 2012).

Measurement faults are defined as rare deterministic errors that cannot be modeled by utilizing zero-mean Gaussian white noise, which might include global positioning system (GPS) clock errors or incorrect data association errors in feature-based localization. If such errors are not detected, they may result in high positioning errors with hazardous consequences. There has been efforts on detecting sensor faults by checking discrepancy among sensor measurements (Kadirkamanathan et al., 2002; Castellanos et al., 2004; Das and Waslander, 2014; Carson et al., 2022; Antonante et al., 2023); however, these techniques lack statistical guarantees on their detection capability. In contrast, this paper utilizes a conservative metric to quantify localization safety: *integrity risk*. Integrity is a metric that certifies each individual system components independently to prove an overall safety level (Kelly and Davis, 1994). Integrity risk is the probability of having large positioning errors without triggering an alarm.

During the past several decades, integrity risk has been the main safety's metric in open-sky GNSS aviation applications (Parkinson and Axelrad, 1988; Brown, 1992; Abuhashim et al., 2010; Joerger et al., 2014; Blanch et al., 2015; Tanil, 2016). Recent efforts have tailored this localization safety metric to mobile robots navigating environments in which GPS is degenerated or obstructed (Mario and Rife, 2010; Li and Waslander, 2019; Hafez et al., 2019; Hassani et al., 2019; Wang et al., 2020; Chen et al., 2020; Antonante et al., 2021; Zhu et al., 2022; Hafez et al., 2024). Recently, the traditional GPS-based integrity monitoring techniques have been extended to state estimators in which the landmark map is gained a priori. The foundation of integrity monitoring for landmark map-based extended Kalman filter's estimator are introduced in Joerger et al. (2016) and Arana et al. (2018) by considering the mis-association risk, namely, the probability of incorrectly associating a feature's measurement that is extracted from a mapped landmark with another landmark, for global and local nearest neighbors criteria, respectively.

For conservativeness, the aforementioned research upper bounds the integrity risk in the case of a data-association error by one hundred percent. To tighten the integrity risk upper bound, Arana et al. (2019b) and Arana (2019) proposed an EKF-based integrity risk quantification method

that can sequentially handle sensor faults from the prior state estimate to the current state estimate while considering the worst conditions, which is improved in Arana et al. (2019a, 2020a) to consider the risk of failure in data association errors and to recursively account for measurement faults within a horizon to tighten of the integrity risk upper bound.

Other research has focused on localization using landmark map-based fixed lag smoothing state estimation algorithm. Arana et al. (2020b) developed a method to monitor integrity offline for a predefined trajectory while utilizing fixed lag smoothing for state estimation and a chi-squared residual for fault detection. Quantifying integrity risk for landmark map-based fixed lag smoothing for state estimation and solution separation for detection is proposed in Hafez et al. (2020b) and compared to the chi-squared residual's approach, which is extended in Hafez et al. (2020a) to account for the mis-association risk of the local nearest neighbors criterion under the impact of prior state's and current measurement's faults.

Despite all the previous research on integrity monitoring for mobile robots, there has been little work on quantifying localization safety for SLAM estimators that jointly estimate a mobile robot's state and a landmark map (Hafez, 2023). Bhamidipati and Gao (2018) tried to quantify the safety Graph-SLAM localizer. However, this paper assumed that the map's landmarks are the actual GPS satellites, which makes the proposed safety technique inapplicable to robotic applications because data association errors, which are the main sources of localization's failure for feature-based state estimators, do not exist among the GPS satellite signals; each GPS satellite signal is augmented with a pseudo code that determines to which satellite the signal corresponds. In addition, this paper has focused only on fault detection and exclusion, which cannot guarantee a predefined level of safety. Other work has proposed an integrity monitoring technique for Graph-SLAM (Bhamidipati and Gao, 2020). However, the method has focused on obtaining the protection level rather than conservatively quantifying mobile robot's localization integrity risk. In addition, it is not clear how the failure probability for each estimated landmark is propagated sequentially, which is what makes SLAM is different that localization. All in all, it is clear that the available literature in monitoring integrity of SLAM is tailored for GPS satellites as landmarks, which makes the form not usable for mobile robots equipped with sensors like lidar or camera; in which the true correspondence of each extracted feature, from raw measurements, to its associated map's landmark is unknown. *Accordingly, this paper will introduce the first conservative approach to quantify mobile robot's localization safety for EKF-SLAM by introducing a recursive technique to evaluate the probability of failure for every estimated landmark, and thus upper bounding the robot's localization integrity risk.* Specifically, the paper's novelties are summarized as follows:

- The bounding up of features' and estimated landmarks' bias impact on their failure probability in EKF-SLAM;

- The derivation of an upper bound on the probability of failure for associated features in EKF-based SLAM;
- The derivation of a sequential method to upper-bound the failure risk of estimated landmarks in EKF-SLAM;
- The development of a practical method to upper-bound the integrity risk for EKF-based SLAM estimator; and
- Detailed study of the relation between data association errors and EKF-based SLAM robot positioning safety.

The remainder of the paper begins with assumptions and a review of local nearest neighbor-based EKF-SLAM where extracted features and estimated landmarks may be affected by unknown systematic errors, or measurement faults, a point that distinguishes this work from others. This is followed by the definition of the utilized localization safety measure; the hazardous misleading information event for the chi-squared fault detector. The distributions of the state estimate error and the fault detector statistic as a function of measurement faults are derived in Section 3. Section 4 defines a practical method to monitor integrity, and Section 5 derives an upper bound on the conditional integrity risk for every failure hypothesis. Section 6 presents the probability of every failure hypothesis as a function of the failure risk for features and landmarks. The probability of data association failure for each extracted feature's measurement from lidar's data and each estimated landmark are derived in Sections 7 and 8. Section 9 assesses the performance of the proposed mobile robot's localization integrity monitor via simulated and experimental scenarios. Finally, the paper's conclusions are presented in Section 10.

2. Assumptions and background

This section demonstrates the assumptions and the necessary background for formulating the localization safety evaluation problem. First, the state evolution's and the measurement's models, along with the assumptions, are presented. Second, the utilized data association criterion for associating features to landmarks, local nearest neighbors algorithm, is defined. Third, the utilized state estimator, Extended Kalman Filter-Simultaneous Localization and Mapping, is detailed. Finally, the utilized robot's localization safety metric is introduced.

2.1. Assumptions

This work assumes the following for evaluating robot safety:

1. the mobile robot estimates its state using EKF-SLAM;
2. the environmental landmark map is unknown a priori;
3. a chi-squared residual is utilized for failure detection;
4. model linearization errors can be upper-bounded via a biased Gaussian uncertainty with inflated covariance;
5. sensor faults are modeled using a single added quantity of lumped additive's and multiplicative's faults; and
6. the hypotheses to be monitored is the ones in which the number of simultaneous faults does not exceed $n_{\max}$.

2.2. State evolution model

This work refers to the current robot's state using $\mathbf{x}_k \in \mathbb{R}^m$, where k is the time epoch and m is the number of robot states to be tracked. The set of tracked states includes robot pose (position and orientation) and the moving biases (unknowns) of sensors like IMU. Moreover, the robot's state evolution is modeled as a nonlinear function of the prior robot state and the control input with an additive process disturbance as, that is:

$$\mathbf{x}_k = \mathbf{g}(\mathbf{x}_{k-1}, \mathbf{u}_{k-1}) + \mathbf{w}_k \quad (1)$$

where $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{W}_k)$ is Gaussian white disturbance, $\mathbf{W}_k$ is a known covariance matrix, $\mathbf{g}(\cdot, \cdot)$ is a known nonlinear function, and $\mathbf{u}_{k-1}$ is the known control comprised of accelerometer, gyroscope, and/or odometer measurements. From here on, to lighten the notation, the time index will be omitted whenever equation's variables have the same index.

2.3. Measurement model

In this paper, the estimated landmarks by EKF-based SLAM, a group of recognizable environmental objects comprised of clusters of characteristic features, are denoted by $\mathbf{l}_r \in \mathbb{R}^{m_F}$; $r = 1, 2, \dots, n_L$ such that r is the time index of the estimated landmark, m_F is the number of features, sensible characteristics of an environmental object, per landmark, and n_L is the number of estimated landmarks at epoch k . In addition, the augmented robot's state vector is referred to as $\mathbf{x}^a = [\mathbf{x}^T \ \mathbf{l}_1^T \dots \ \mathbf{l}_{n_L}^T]^T \in \mathbb{R}^{m+n_L m_F}$. The measurements, a set of decoded sensors' readings that are sensitive to physical environmental characteristics, vector, $\mathbf{z} \in \mathbb{R}^n$, can be modeled as a nonlinear function of the robot's state and the estimated landmarks with an additive measurement noise and fault, that is:

$$\mathbf{z} = \mathbf{h}(\mathbf{x}^a) + \mathbf{v} + \mathbf{f} \quad (2)$$

where n refers to the total number of sensor measurements, $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{V}) \in \mathbb{R}^n$ is a Gaussian white sensor measurement noise, $\mathbf{V}$ is the known noise covariance matrix, $\mathbf{h}(\cdot, \cdot)$ is a known nonlinear sensor measurement function, and $\mathbf{f} \in \mathbb{R}^n$ is the sensor measurement faults vector, which is $\mathbf{0}$ if and only if all sensor measurements are non-faulted. To simplify notation, we assume that the sensor measurement's vector is comprised of only the extracted features' measurements $\mathbf{z}_i \in \mathbb{R}^{m_F}$, measurements that their corresponding landmarks have to be specified in order for their measurements' models to be specified, without including non-feature measurements, measurements that their measurements' models are specified without correspondence selection. Thus, assuming that only n_a^F out of the n^F extracted features have been associated to the estimated landmarks, the size of the sensor measurement vector $\mathbf{z} = [\mathbf{z}_1^T \ \dots \ \mathbf{z}_{n_a^F}^T]^T$ will be

$n = n_a^F m_F$. This work models faults as unknown deterministic errors to account for the (non-Gaussian distributed) rarely occurring failure events in SLAM that occur when an extracted feature measurement is incorrectly associated to an estimated landmark or when a new landmark is added from a feature that corresponds to a pre-estimated landmark by EKF-SLAM or a moving object.

2.4. Local nearest neighbors

The data association algorithm is responsible for associating the extracted features to the estimated landmarks. In mobile robotics, there are two main data association criteria: global nearest neighbors and local nearest neighbors. Compared to global nearest neighbors where the correspondences of all extracted features are estimated instantaneously, local nearest neighbors estimates the correspondence of each extracted feature independently; making it more computationally efficient (Bar-Shalom et al., 2004). Accordingly, the local nearest neighbors algorithm is utilized for data association. In this work, the individual innovation vector between the i^{th} feature measurement, $\mathbf{z}_i$, and the r^{th} landmark's predicted estimate, $\bar{\mathbf{l}}_r$, is utilized as a proximity measure, $\gamma_{i,r}$, as follows, that is:

$$\gamma_{i,r} = \mathbf{z}_i - \mathbf{h}_r(\bar{\mathbf{x}}^a) \quad \text{with} \quad \mathbf{Y}_i = \mathbf{H}_{i*} \bar{\mathbf{P}}^a \mathbf{H}_{i*}^T + \mathbf{V}_i \quad (3)$$

as its covariance, $\mathbf{V}_i \in \mathbb{R}^{m_F}$ is the noise covariance of the i^{th} feature, $\bar{\mathbf{P}}^a$ is the covariance for the prediction estimate error of the augmented state, $\delta \bar{\mathbf{x}}^a = \mathbf{x}^a - \bar{\mathbf{x}}^a$, i^* is the index of the landmark from which the i^{th} feature's measurement is extracted, and $\mathbf{H}_{i*} = \partial \mathbf{h}_{i*} / \partial \mathbf{x}^a|_{\bar{\mathbf{x}}^a}$ is the Jacobian of the sensor's measurement function for landmark $\mathbf{l}_{i*}$. Note that the columns of $\mathbf{H}_{i*}$ are zero vectors' set except for $\mathbf{x}$ and $\mathbf{l}_{i*}$.

2.4.1. Associating features to estimated landmarks. In the local nearest neighbors' data association criterion, each of the n^F features, $\mathbf{z}_i$, is associated to the closest estimated landmark, $\mathbf{l}_{l_i}$ in the Mahalanobis space if and only if the estimated landmark $\mathbf{l}_{l_i}$ is not extremely far as follows, that is:

$$\|\gamma_{i,l_i}\|_{\mathbf{Y}_i^{-1}} < T_a \quad \text{where} \quad l_i = \underset{r=1:n_{k-1}^L}{\operatorname{argmin}} \|\gamma_{i,r}\|_{\mathbf{Y}_i^{-1}}, \quad (4)$$

T_a is the separation threshold that reduces the incorrect data association rate, and $\|\mathbf{a}\|_{\mathbf{B}} = \sqrt{\mathbf{a}^T \mathbf{B} \mathbf{a}}$ is the weighted ℓ^2 -norm. Note that if $l_i = i^*$, then the i^{th} feature measurement is correctly associated, and if none of the landmarks satisfies (4), the i^{th} extracted feature measurement is not associated.

2.4.2. Using features to add new landmarks. In local nearest neighbors, the $n^F - n_a^F$ non-associated features are utilized to add new estimated landmarks. Specifically, a new estimated landmark is added from the j^{th} non-associated features, $\mathbf{z}_j$, if it is far from all the n_{k-1}^L estimated landmarks:

$$\bigcap_{r=1}^{n_{k-1}^L} \|\gamma_{j,r}\|_{\mathbf{Y}_j^{-1}} > T_n \quad (5)$$

where $T_a \leq T_n$ and T_n is a landmark separation threshold that reduces the rate of adding estimated landmarks from feature measurements that correspond to pre-estimated landmarks. Assuming that n_n^F estimated landmarks are added, then the remaining $n^F - n_a^F - n_n^F$ feature measurements are discarded.

2.5. Extended Kalman filter-based SLAM

EKF-SLAM is an online SLAM estimator that incrementally builds a map of estimated landmarks while localizing the mobile robot. This algorithm can be decomposed into three steps: prediction step, data association step, and update step (Thrun, 2002). A brief description is given here to highlight the notations utilized in the following sections of the paper.

2.5.1. Prediction step. The state propagation model (1) is utilized to get a predicted estimate of the augmented mobile robot's state vector at the next epoch, $\bar{\mathbf{x}}_k^a$, with its covariance, $\bar{\mathbf{P}}_k^a$, from the prior estimate of the augmented mobile robot's state vector, $\hat{\mathbf{x}}_{k-1}^a$, and its covariance matrix, $\hat{\mathbf{P}}_{k-1}^a$ as follows:

$$\bar{\mathbf{x}}_k^a = \begin{bmatrix} \bar{\mathbf{x}}_k^T & \bar{\mathbf{l}}_1^T & \dots & \bar{\mathbf{l}}_{n_{k-1}^L}^T \end{bmatrix}^T = \begin{bmatrix} \mathbf{g}(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_{k-1})^T & \hat{\mathbf{l}}_1^T & \dots & \hat{\mathbf{l}}_{n_{k-1}^L}^T \end{bmatrix}^T \quad (6)$$

$$\bar{\mathbf{P}}_k^a = \mathbf{G}_{k-1}^a \hat{\mathbf{P}}_{k-1}^a \mathbf{G}_{k-1}^{aT} + \mathbf{W}_k^a \quad (7)$$

where $\mathbf{G}_{k-1}^a$ and $\mathbf{W}_{k-1}^a$ are the augmented formulations of the state evolution Jacobian matrix and the process disturbance covariance matrix, respectively, which are defined as follows:

$$\mathbf{G}_{k-1}^a = \begin{bmatrix} \mathbf{G}_{k-1} & \mathbf{0}_{m \times n_{k-1}^L m_F} \\ \mathbf{0}_{n_{k-1}^L m_F \times m} & \mathbf{I}_{n_{k-1}^L m_F \times n_{k-1}^L m_F} \end{bmatrix} \quad (8)$$

$$\mathbf{W}_{k-1}^a = \begin{bmatrix} \mathbf{W}_{k-1} & \mathbf{0}_{m \times n_{k-1}^L m_F} \\ \mathbf{0}_{n_{k-1}^L m_F \times m} & \mathbf{0}_{n_{k-1}^L m_F \times n_{k-1}^L m_F} \end{bmatrix} \quad (9)$$

where $\mathbf{G}_{k-1} = \partial \mathbf{g} / \partial \mathbf{x}|_{\hat{\mathbf{x}}_{k-1}}$ is the Jacobian's matrix of the mobile robot's state evolution function, and $\mathbf{I}_\infty$ is an identity matrix. Note that when EKF-SLAM estimation algorithm is initialized, the augmented mobile robot state vector, $\bar{\mathbf{x}}_k^a$, will only be composed of mobile robot's state vector ($n_{k-1}^L = 0$).

2.5.2. Data association. The extracted features' measurements, $\mathbf{z}_i, \forall i = 1, \dots, n_k^F$, are associated with the pre-estimated landmarks, $\bar{\mathbf{l}}_r, \forall r = 1, \dots, n_{k-1}^L$, by using the predicted state estimate, $\bar{\mathbf{x}}$, see (4). In addition, the non-associated feature measurements are tested whether to be

added as new landmarks (at the end of update step) or to be discarded, see (5).

2.5.3. Update step. The associated features' measurements along with its corresponding pairs of estimated landmarks and the predicted state will be used to build the innovation:

$$\begin{aligned} \gamma &= \begin{bmatrix} \mathbf{z}_1^T - \mathbf{h}_{l_1}(\bar{\mathbf{x}}^a)^T & \cdots & \mathbf{z}_{n_n^F}^T - \mathbf{h}_{l_{n_n^F}}(\bar{\mathbf{x}}^a)^T \end{bmatrix}^T \\ &= \mathbf{z} - \mathbf{h}(\bar{\mathbf{x}}^a) \end{aligned} \quad (10)$$

The difference between the sensor measurements and the predicted measurements of estimated mobile robot state and landmarks, which is captured by the innovation vector, γ , is utilized to get an updated estimate of the augmented robot state, $\hat{\mathbf{x}}^a$, and its state covariance matrix, $\hat{\mathbf{P}}^a$, as follows, that is:

$$\hat{\mathbf{x}}^a = \bar{\mathbf{x}}^a + \mathbf{L}\gamma \quad (11)$$

$$\hat{\mathbf{P}}^a = (\mathbf{I}_{m \times m} - \mathbf{L}\mathbf{H})\bar{\mathbf{P}}^a \quad (12)$$

where $\mathbf{L}$ and $\mathbf{H}$ are the matrices of Kalman's gain and the measurement's Jacobian, respectively, which are defined as follows:

$$\mathbf{L} = \bar{\mathbf{P}}^a \mathbf{H}^T \mathbf{Y}^{-1} \quad (13)$$

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{l_1}^T & \cdots & \mathbf{H}_{l_{n_n^F}}^T \end{bmatrix}^T \quad (14)$$

where $\mathbf{Y}$ is the covariance matrix of the innovation's vector:

$$\mathbf{Y} = \mathbf{H}\bar{\mathbf{P}}^a \mathbf{H}^T + \mathbf{V} \quad (15)$$

2.5.4. Adding new landmarks. Finally, the update's step is completed by adding new landmarks, from non-associated features that passed the test in (5). This is done by projecting each of these n_n^F non-associated features on the landmark's space by utilizing the predicted robot's state estimate, $\bar{\mathbf{x}}^a$, that is:

$$\hat{\mathbf{l}}_j = \Phi(\bar{\mathbf{x}}, \mathbf{z}_j), \forall j = 1, \dots, n_n^F \quad (16)$$

$$\hat{\mathbf{R}}_j = \Phi_{j,x} \bar{\mathbf{P}} \Phi_{j,x}^T + \Phi_{j,z} \mathbf{V}_j \Phi_{j,z}^T \quad (17)$$

where $\hat{\mathbf{l}}_j$ is the added landmark from the j^{th} feature with $\hat{\mathbf{R}}_j$ as its covariance matrix, and $\Phi(\cdot, \cdot)$ is a mapping function from the feature space to the landmark space, with $\Phi_{j,x} = \partial\Phi/\partial\mathbf{x}|_{\bar{\mathbf{x}}, \mathbf{z}_j}$ and $\Phi_{j,z} = \partial\Phi/\partial\mathbf{z}^j|_{\bar{\mathbf{x}}, \mathbf{z}_j}$ as its Jacobian matrices with respect to mobile robot's state and the mapped feature measurement, respectively. Thus, the updated estimate of the augmented state's vector and its covariance will become as:

$$\hat{\mathbf{x}}^a = \begin{bmatrix} \hat{\mathbf{x}}^a & \hat{\mathbf{l}}_1^T & \cdots & \hat{\mathbf{l}}_{n_n^F}^T \end{bmatrix}^T \quad (18)$$

$$\hat{\mathbf{P}}^a = \begin{bmatrix} \hat{\mathbf{P}}^a & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{R}}_{l_1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \mathbf{0} \\ \mathbf{0} & \cdots & \mathbf{0} & \hat{\mathbf{R}}_{l_{n_n^F}} \end{bmatrix} \quad (19)$$

where $\hat{\mathbf{x}}^a$ and $\hat{\mathbf{P}}^a$ will serve as the set of inputs to the mobile robot's EKF-based SLAM prediction step at the subsequent time epoch, and the state estimation algorithm will re-curse.

In the next section, the description of a localization failure event, *Hazardous Misleading Information*, will be presented.

2.6. Hazardous misleading information

This paper models the failure of EKF-based SLAM using the Hazardous Misleading Information (HMI) event, which happens when the estimate error in the state-of-interest, for example, car's lateral error in automotive applications, surpasses an acceptable limit and the failure detection's mechanism does not trigger an alarm (RTCA Special Committee 159, 2004):

$$HMI = |\alpha^T \delta \hat{\mathbf{x}}^a| > l \cap q \leq T \quad (20)$$

where $\delta \hat{\mathbf{x}}^a = \mathbf{x}^a - \hat{\mathbf{x}}^a$ is the update step estimate error of the augmented robot state vector, $\alpha \in \mathbb{R}^{m+n_{k-1}^L m_F}$ is the state-of-interest extraction vector, l is the predefined alert-limit, q is the fault detector's statistic, and T is its predefined threshold.

The fault detector is a statistical measure of inconsistency in sensor measurements that triggers an alarm whenever it is greater than T_k . This work uses the standardized innovation's squared norm as the fault detector (Arana et al., 2019b), that is:

$$q = \|\gamma\|_{\mathbf{Y}^{-1}}^2 \quad (21)$$

Integrity risk is defined as the HMI event's probability, $P(HMI)$. To quantify such a metric, the distributions of the fault detector and the state-of-interest's estimate error have to be derived, which will be done next; Figure 1 serves as a guide.

3. Estimate error and detector distributions

This section addresses the impact of sensor's measurement faults on the distributions of EKF-SLAM mobile robot state estimates errors and the fault detector. Since EKF-SLAM is a recursive estimator, sensor measurement's faults at any time epoch will not only bias the EKF-SLAM mobile robot state's estimate at that specific epoch but also in all future epochs.

3.1. Prior state estimate error distribution

The estimate error's distribution of EKF-SLAM's augmented prior robot's state can be modeled as a non-zero

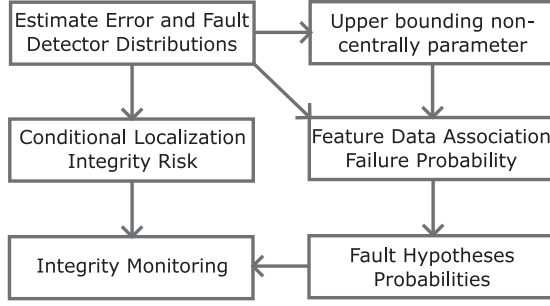


Figure 1. The flowchart illustrates the sequential dependencies between the sections of the paper starting from three until eight.

expectation's Gaussian distribution with a known covariance matrix, $\hat{\mathbf{P}}_{k-1}^a$:

$$\delta \hat{\mathbf{x}}_{k-1}^a = \mathbf{x}_{k-1}^a - \hat{\mathbf{x}}_{k-1}^a \sim \mathcal{N}(\hat{\mathbf{f}}_{k-1}^a, \hat{\mathbf{P}}_{k-1}^a) \quad (22)$$

$$\hat{\mathbf{f}}_{k-1}^a = \begin{bmatrix} \hat{\mathbf{f}}_{x_{k-1}}^T & \hat{\mathbf{f}}_{l_1}^T \dots & \hat{\mathbf{f}}_{l_{n_{k-1}^L}}^T \end{bmatrix}^T \quad (23)$$

where $\hat{\mathbf{f}}_{k-1}^a$ and $\hat{\mathbf{f}}_{x_{k-1}}$ are the fault vectors of the augmented prior state's estimate and the prior robot's state estimate, respectively, resulting from the utilization of faulted sensor's measurements in the robot's state estimation steps at former time epochs, and $\hat{\mathbf{f}}_{l_i}$ is the fault vector in the i^{th} estimated landmark resulting from incorrectly associating it with non-corresponding feature measurements at former time epochs.

3.2. Predicted state estimate error distribution

The prior robot state faults will bias the state estimate at the next epoch using the prediction step. Specifically, EKF-SLAM predicted state estimate error's distribution is gained by linearizing the state evolution model at the prior estimate:

$$\begin{aligned} \delta \bar{\mathbf{x}}_k^a &= \mathbf{x}_k^a - \bar{\mathbf{x}}_k^a \\ &= \left[\mathbf{g}(\mathbf{x}_{k-1}, \mathbf{u})^T - \mathbf{g}(\hat{\mathbf{x}}_{k-1}, \mathbf{u})^T + \mathbf{w}_k^T \quad \delta \hat{\mathbf{l}}_1^T \quad \dots \quad \delta \hat{\mathbf{l}}_{n_{k-1}^L}^T \right]^T \\ &\approx \left[\delta \hat{\mathbf{x}}_{k-1}^T \mathbf{G}_{k-1}^T + \mathbf{w}_k^T \quad \delta \hat{\mathbf{l}}_1^T \quad \dots \quad \delta \hat{\mathbf{l}}_{n_{k-1}^L}^T \right]^T \\ &\approx \mathbf{G}_{k-1}^a \delta \hat{\mathbf{x}}_{k-1}^a + \left[\mathbf{w}_k^T \quad \mathbf{0}_{1 \times n_{k-1}^L m_F} \right]^T \\ &\sim \mathcal{N} \left(\underbrace{\mathbf{G}_{k-1}^a \hat{\mathbf{f}}_{k-1}^a}_{=\hat{\mathbf{f}}_k^a}, \hat{\mathbf{P}}_k^a \right) \end{aligned} \quad (24)$$

where $\hat{\mathbf{f}}_k^a$ is the augmented prediction state fault vector, and $\hat{\mathbf{P}}_k^a$ is the prediction estimate error's covariance defined in (7).

3.3. Updated state estimate error distribution

From (11), the update step's state estimate, $\hat{\mathbf{x}}^a$, is gained by optimally fusing the prediction step's state estimate, $\bar{\mathbf{x}}^a$, with

the measurements, $\mathbf{z}$. Thus, the estimate error's distribution of EKF-SLAM's update step is affected by both faults in the prediction step's estimate, $\bar{\mathbf{f}}^a$, and in the measurements, $\mathbf{f}$, that is:

$$\begin{aligned} \delta \hat{\mathbf{x}}^a &= \mathbf{x}^a - \hat{\mathbf{x}}^a \\ &= \delta \bar{\mathbf{x}}^a - \mathbf{L} \gamma \\ &= \delta \bar{\mathbf{x}}^a - \mathbf{L}(\mathbf{z} - \mathbf{h}(\bar{\mathbf{x}}^a)) \\ &= \delta \bar{\mathbf{x}}^a - \mathbf{L}(\mathbf{h}(\mathbf{x}^a) - \mathbf{h}(\bar{\mathbf{x}}^a) + \mathbf{v} + \mathbf{f}) \\ &\approx \delta \bar{\mathbf{x}}^a - \mathbf{L}(\mathbf{H} \delta \bar{\mathbf{x}}^a + \mathbf{v} + \mathbf{f}) \\ &\approx (\mathbf{I}_{m \times m} - \mathbf{LH}) \delta \bar{\mathbf{x}}^a - \mathbf{L}(\mathbf{v} + \mathbf{f}) \\ &\sim \mathcal{N} \left(\underbrace{[(\mathbf{I}_{m \times m} - \mathbf{LH}) \quad -\mathbf{L}]}_{=\mathbf{A}} \underbrace{\begin{bmatrix} \bar{\mathbf{f}}^a \\ \mathbf{f} \end{bmatrix}}_{=\check{\mathbf{f}}}, \hat{\mathbf{P}}^a \right) \end{aligned} \quad (25)$$

where $\check{\mathbf{f}}$ is the combined predicted estimate's and measurements' fault vector with $\mathbf{A}$ as its projection matrix on the updated estimate, and $\hat{\mathbf{P}}^a$ is the covariance of the updated estimate error, check (12). Given the distribution of $\delta \hat{\mathbf{x}}^a$, the state-of-interest's estimate error distribution is expressed as:

$$\alpha^T \delta \hat{\mathbf{x}}^a \sim \mathcal{N}(\alpha^T \mathbf{A} \check{\mathbf{f}}, \alpha^T \hat{\mathbf{P}}^a \alpha) \quad (26)$$

3.4. Innovation vector distribution

Before introducing the detector distribution, the innovation distribution has to be derived. Recall that the innovation is a discrepancy between the actual and predicted measurement. Thus, similar to the updated estimate error, its distribution is a function of the predicted estimate's and sensor's faults, that is:

$$\begin{aligned} \gamma &= \mathbf{z} - \mathbf{h}(\bar{\mathbf{x}}^a) \\ &= \mathbf{h}(\mathbf{x}^a) - \mathbf{h}(\bar{\mathbf{x}}^a) + \mathbf{v} + \mathbf{f} \\ &\approx \mathbf{H} \delta \bar{\mathbf{x}}^a + \mathbf{v} + \mathbf{f} \\ &\sim \mathcal{N} \left(\underbrace{[\mathbf{H} \quad \mathbf{I}_{n \times n}]}_{=\mathbf{B}} \check{\mathbf{f}}, \mathbf{Y} \right) \end{aligned} \quad (27)$$

where $\mathbf{B}$ is the projection matrix of the prediction step state estimate's and measurement's faults on the innovation vector, and $\mathbf{Y}$ is the innovation's covariance matrix, as shown in (15).

3.5. Fault detector statistic distribution

From (21), the detector's statistic is the squared norm of a Gaussian random vector weighted by its covariance matrix inverse. Thus, it can be represented as the squared norm of a non-zero mean standard Gaussian vector, $\mathbf{Y}^{-1/2} \gamma$, making its distribution a non-central chi-squared

with n degrees of freedom and λ as a non-centrality [Arana et al. \(2019b\)](#), that is:

$$\begin{aligned} q &= \|\mathbf{Y}\|_{\mathbf{Y}^{-1}}^2 \\ &= \|\mathbf{Y}^{-1/2}\mathbf{Y}\|^2; \text{ where } \mathbf{Y}^{-1/2}\mathbf{Y} \sim \mathcal{N}(\mathbf{Y}^{-1/2}\mathbf{B}\check{\mathbf{f}}, \mathbf{I}_{n \times n}) \quad (28) \\ &\sim \chi_{n,\lambda}^2, \quad \text{with } \lambda = \check{\mathbf{f}}^T \mathbf{B}^T \mathbf{Y}^{-1} \mathbf{B} \check{\mathbf{f}} \end{aligned}$$

where $\chi_{a,b}^2$ is a non-central chi-squared distribution with a as degrees of freedom and b as a non-centrality's parameter. Note that having relatively large sensors measurements' or prediction step state's faults will increase the non-centrality parameter and so will shift the chi-squared distribution to the right, making the detection probability larger, and vice versa.

Given the distribution of fault detector's statistic, its safety threshold, T , is chosen such that the detection risk in the non-faulted condition (i.e., false alarms, $P(q > T; \check{\mathbf{f}} = \mathbf{0})$) is bounded by a predefined value, I_{FA} . By utilizing the fact that in the non-faulted case, the detector distribution reduces to a central chi-squared distribution, the fault detector's threshold is computed as [Joerger et al. \(2014\)](#) and [Arana et al. \(2019a\)](#), that is:

$$T = X_n^{-2}[1 - I_{FA}] \quad (29)$$

where $X_a^{-2}[\cdot]$ is the inverse Cumulative Distribution Function (CDF) of the central chi-squared distribution. The following section uses the distributions of the EKF-SLAM's estimate error and failure detector to quantify the localization's safety.

4. Integrity monitoring

This section introduces the proposed technique to monitor mobile robot's localization safety for EKF-SLAM, where the landmark map is not known a priori. Recall that this paper uses the integrity risk as a safety metric, which is defined as:

$$P(HMI) = P(|\alpha^T \delta \hat{\mathbf{x}}^a| > l \cap q \leq T) \quad (30)$$

Evaluating the joint's probability of a Gaussian and a chi-squared distribution is so complex. However, [Joerger et al. \(2014\)](#) proved that the state estimate error and the chi-squared fault detector are independent random variables as follows:

$$\begin{aligned} P(HMI) &= P(|\alpha^T \delta \hat{\mathbf{x}}^a| > l) P(q \leq T) \\ &= \left(\Phi \left[\frac{-l - \alpha^T \mathbf{A} \check{\mathbf{f}}}{\alpha^T \hat{\mathbf{P}}^a \alpha} \right] + 1 - \Phi \left[\frac{l - \alpha^T \mathbf{A} \check{\mathbf{f}}}{\alpha^T \hat{\mathbf{P}}^a \alpha} \right] \right) X_{n,\lambda}^2[T] \end{aligned} \quad (31)$$

where $\Phi[\cdot]$ is the CDF of standard Gaussian's distribution, and $X_{a,b}^2$ is the CDF of non-central chi-squared distribution with a as degrees of freedom and b as a non-centrality. Note that the integrity risk is a function of the unknown combined

prediction step estimate's and sensors measurement's fault, $\check{\mathbf{f}}$. Accordingly, the integrity risk is quantified under mutually exclusive and collectively exhaustive set of fault hypotheses, $\{H^0, H^1, \dots, H^{n_H}\}$; with H^0 as the fault-free hypothesis, that is:

$$P(HMI_k) = \sum_{h=0}^{n_H} \underbrace{P(HMI | H^h)}_a \underbrace{P(H^h)}_b \quad (32)$$

where $P(HMI | H^h)$ is the integrity risk given the h^{th} fault hypothesis, and $P(H^h)$ is the h^{th} fault hypothesis probability.

Each fault hypothesis specifies which set of measurements are faulted, but does not specify the quantity of the fault, for example:

$$\text{Under } H_k^h: \quad \mathbf{E}_h \check{\mathbf{f}} \neq \mathbf{0} \quad (33)$$

where $\mathbf{E}_h$ is the faulted measurements extraction matrix for the h^{th} hypothesis. As an example, if the number of extracted features is $n_{a_k}^F = 4$ and the number of estimated landmarks is $n_{k-1}^L = 5$, with the h^{th} fault hypothesis marking the fourth feature and the first landmark as faulted, then the faulted measurements extraction matrix, $\mathbf{E}_h$, is expressed as follows:

$$\mathbf{E}_h = \begin{bmatrix} \mathbf{0}_{m_F \times 3m_F} & \mathbf{I}_{m_F \times m_F} & \mathbf{0}_{m_F \times m} & \mathbf{0}_{m_F \times m_F} & \mathbf{0}_{m_F \times 4m_F} \\ \mathbf{0}_{m_F \times 3m_F} & \mathbf{0}_{m_F \times m_F} & \mathbf{0}_{m_F \times m} & \mathbf{I}_{m_F \times m_F} & \mathbf{0}_{m_F \times 4m_F} \\ \mathbf{0}_{m \times 3m_F} & \mathbf{0}_{m \times m_F} & \mathbf{I}_{m \times m} & \mathbf{0}_{m \times m_F} & \mathbf{0}_{m \times 4m_F} \end{bmatrix} \quad (34)$$

where the last row extracts the predicted estimate fault to account for the impact of feature faults at past epochs on the prior estimate, $\hat{\mathbf{x}}_{k-1}$, which results from the recursive nature of EKF-based SLAM. To get a conservative safety measure, the next sections will upper-bound $P(HMI | H^h)$ and $P(H^h)$.

5. Conditional integrity risk

This section presents a conservative technique to quantify the conditional mobile robot's localization integrity risk given the fault hypothesis; term a in (32). Because the actual fault vector, $\check{\mathbf{f}}$, is unknown, $P(HMI | H^h)$ can be upper-bounded by replacing $\check{\mathbf{f}}$ with the worst-case fault vector, $\check{\mathbf{f}}_h^w$, that is, the fault vector that maximizes the conditional integrity risk, that is:

$$\check{\mathbf{f}}_h^w = \check{\mathbf{f}}_h^{dir} \check{\mathbf{f}}_h^{mag} \quad (35)$$

where $\check{\mathbf{f}}_h^{dir}$ and $\check{\mathbf{f}}_h^{mag}$ are the h^{th} hypothesis worst-case fault vector's direction (unit vector) and magnitude, respectively.

[Joerger et al. \(2014\)](#) and [Angus \(2006\)](#) demonstrated that the fault vector direction that maximizes the conditional robot's localization integrity risk is the same as the one that maximizes its failure mode slope, s_h , which is defined as follows, that is:

$$s_h = \frac{\mathbb{E}[\alpha^T \delta \hat{\mathbf{x}}^a | H_k^h]^T \mathbb{E}[\alpha^T \delta \hat{\mathbf{x}}^a | H_k^h]}{\lambda[q | H^h]} \quad (36)$$

where $\mathbb{E}[\cdot]$ is the expectation vector of a random variable, and $\lambda[\cdot]$ is the non-centrality's parameter of non-central chi-squared distribution. By substituting (26) and (28) back in (36), the failure mode slope can be expressed as follows, that is:

$$s_h = \frac{\tilde{\mathbf{f}}_h^T \mathbf{E}_h^T \mathbf{E}_h \mathbf{A}^T \alpha \alpha^T \mathbf{A} \mathbf{E}_h^T \tilde{\mathbf{f}}_h}{\tilde{\mathbf{f}}_h^T \mathbf{E}_h^T \mathbf{E}_h \mathbf{B}^T \mathbf{Y}^{-1} \mathbf{B} \mathbf{E}_h^T \tilde{\mathbf{f}}_h} \quad (37)$$

where $\mathbf{E}_h^T \mathbf{E}_h$ is included to zero the non-faulted measurements in $\tilde{\mathbf{f}}_h$ given the h^{th} hypothesis. An analytical solution for the worst case vector direction is proven in Joerger et al. (2014), which can be written in our EKF-SLAM problem as follows:

$$\tilde{\mathbf{f}}_h^{\text{dir}} = \mathbf{E}_h^T (\mathbf{E}_h \mathbf{B}^T \mathbf{Y}^{-1} \mathbf{B} \mathbf{E}_h^T)^{-1} \mathbf{E}_h \mathbf{A}^T \alpha \quad (38)$$

Given the worst-case fault vector's direction, its magnitude, $\tilde{\mathbf{f}}_h^{\text{mag}}$, is obtained using line search (Joerger et al., 2014), that is:

$$\begin{aligned} \tilde{\mathbf{f}}_h^{\text{mag}} &= \underset{\mathbf{f}}{\operatorname{argmax}} [P(HMI | H^h)] \\ &= \underset{\mathbf{f}}{\operatorname{argmax}} \left[\left(\Phi \left[\frac{-l - \alpha^T \mathbf{A} \tilde{\mathbf{f}}_h^{\text{dir}}}{\alpha^T \hat{\mathbf{P}} \alpha} \right] + 1 - \Phi \left[\frac{l - \alpha^T \mathbf{A} \tilde{\mathbf{f}}_h^{\text{dir}}}{\alpha^T \hat{\mathbf{P}} \alpha} \right] \right) X_{n, \lambda_{n, \tilde{\mathbf{f}}_h^{\text{dir}}}}^2 [T] \right], \text{where:} \end{aligned} \quad (39)$$

$$\lambda_{n, \tilde{\mathbf{f}}_h^{\text{dir}}}^{\text{dir}} = \tilde{\mathbf{f}}_h^{\text{dir}^T} \mathbf{B}^T \mathbf{Y}^{-1} \mathbf{B} \tilde{\mathbf{f}}_h^{\text{dir}} \quad (40)$$

This section presented an upper bound on the conditional integrity risk given the failure hypothesis. In the next section, the hypothesis probability, term b in (32), will be quantified.

6. Fault hypotheses probabilities

This section derives the probability of each fault hypothesis, $P(H^h)$. Recall from (25), that there are three types of faults:

1. prior mobile robot's state estimate faults, $\hat{\mathbf{f}}_{x_{k-1}}$;
2. estimated mapped landmark's faults, $\hat{\mathbf{f}}_l$; and
3. extracted feature measurement's faults, $\mathbf{f}_k^i$.

We assume that the prior robot's state estimate is always faulted, $\hat{\mathbf{f}}_{x_{k-1}} \neq \mathbf{0}$, because its failure risk approaches one in a short amount of time, for example, using a single faulted measurement at any time epoch makes it faulted. Given that in the fault hypothesis H_k^h r^F and r^L associated features and landmarks are faulted ($r^L \leq r^F$), $P(H_k^h)$ can be expressed as follows, that is:

$$P(H_k^h) = P \left(\bigcap_{i=1}^{r^F} F_k^{z_i} \bigcap_{i=r^F+1}^{n_{a_k}^F} \neg F_k^{z_i} \bigcap_{i=1}^{r^L} F_{k-1}^{l_i} \bigcap_{i=r^L+1}^{n_{a_k}^F} \neg F_{k-1}^{l_i} \right) \quad (41)$$

where $F_k^{z_i}$ refers to the failure event of the i^{th} associated feature with $F_{k-1}^{l_i}$ as the failure event of its corresponding estimated landmark, and $\neg$ refers to the inversion operator of a Boolean event. Using the generic independence assumption among the failure events of the associated features and the estimated landmarks, in the map, the probability of the h^{th} fault hypothesis, in (41), can be upper-bounded as follows:

$$\begin{aligned} P(H_k^h) &\leq \prod_{i=1}^{n_F} P(F_k^{z_i}) \prod_{i=n_F+1}^{n_{a_k}^F} (1 - P(F_k^{z_i})) \prod_{i=1}^{r^L} P(F_{k-1}^{l_i}) \\ &\quad \prod_{i=r^L+1}^{n_{a_k}^F} (1 - P(F_{k-1}^{l_i})) \end{aligned} \quad (42)$$

A special case of $P(H_k^h)$ is $P(H_k^0)$, which is expressed as:

$$P(H_k^0) \leq \prod_{i=1}^{n_{a_k}^F} (1 - P(F_k^{z_i})) \prod_{j=1}^{n_{a_k}^F} (1 - P(F_{k-1}^{l_j})) \quad (43)$$

Quantifying the integrity risk for the entire potential fault hypotheses can be computationally expensive, even for few measurements; as its time complexity can be proven to be exponential with the number of feature measurements, that is, $\mathcal{O}(n^n)$ in big-O notation. Therefore, this paper upper bounds the integrity risk by only monitoring the fault hypotheses in which the number of simultaneous measurement faults is less than or equal to a specific number, $n_{\max}$, while accounting for the risk of exceeding $n_{\max}$ simultaneous faults. At the expense of loosening the integrity risk's upper bound, this bound reduces the time complexity to become polynomial with the number of feature measurements, that is, $\mathcal{O}(n^{n_{\max}})$ in big-O notation. Thus, this reduction, in the number of fault hypotheses to be monitored, results in a trade-off between the computing time and the over-confidence of the upper bound. Accordingly, the upper bound on the mobile robot's localization integrity risk can be expressed as follows, that is:

$$\begin{aligned} P(HMI) &\leq I_{>n_{\max}} + \sum_{\substack{h=0 \\ h \notin >n_{\max}}}^{n_H} P(HMI | H^h) P(H^h) (1 - I_{>n_{\max}}) \\ &\leq I_{>n_{\max}} + \sum_{\substack{h=0 \\ h \notin >n_{\max}}}^{n_H} P(HMI | H^h) P(H^h) \end{aligned} \quad (44)$$

where $I_{>n_{\max}}$ is the event's probability of having more than $n_{\max}$ simultaneous sensor measurement faults, which can be upper-bounded by using appendix C in Blanch et al. (2012):

$$I_{>n_{\max}} \leq \frac{\left(\sum_{i=1}^{n_{a_k}^F} P(F_k^{z_i}) + \sum_{j=1}^{n_{a_k}^F} P(F_{k-1}^{l_j}) \right)^{n_{\max}+1}}{(n_{\max}+1)!} \quad (45)$$

In the following section, the probability of failure for each extracted feature and estimated landmark will be introduced.

7. Data association failure probability

This section derives the probability of failure for the associated features measurements and the estimated landmarks.

7.1. Feature's failure probability

The failure modes in associating features and landmarks are:

1. *Mis-association* – occurs when a feature measurement that corresponds to a pre-estimated landmark is associated to a non-corresponding estimated landmark.
2. *Unmapped-association* – occurs when a feature measurement that corresponds to an unestimated landmark or moving object is associated to an estimated landmark.

The i^{th} associated feature's failure probability is defined as:

$$P(F^{z_i}) = P(MA_i \cup UA_i) \quad (46)$$

where MA_i and UA_i are the mis-association and unmapped-association failure events of the i^{th} associated feature, $\mathbf{z}_i$, respectively. Because MA_i and UA_i are disjoint events, that is, cannot occur for the same feature, $P(F^{z_i})$ can be written as:

$$P(F^{z_i}) = P(MA_i) + P(UA_i) \quad (47)$$

This paper assumes $P(UA_i)$ a constant that is upper-bounded empirically by using parameter identification under different environmental conditions (Chen et al., 2024); the detailed focus is shed on deriving $P(MA_i)$. Given that the estimated landmark, that the i^{th} feature is associated to, is known, $\mathbf{l}_i$, MA_i occurs when the features corresponding to the nearby landmarks, $\{\mathbf{z}_j : j \neq i\}$, is associated to the landmark $\mathbf{l}_i$, that is:

$$P(MA_i) =$$

$$P\left(\bigcup_{\substack{j=1 \\ j \neq i}}^{n_{L_{k-1}}^{FoV}} \underbrace{\left\| \mathbf{y}_{j,i} \right\|_{\mathbf{Y}_j^{-1}} < T_a}_a \underbrace{\bigcap_{\substack{r=1 \\ r \neq j}}^{n_{L_{k-1}}^{FoV}} \left\| \mathbf{y}_{j,i} \right\|_{\mathbf{Y}_j^{-1}} \leq \left\| \mathbf{y}_{r,i} \right\|_{\mathbf{Y}_j^{-1}}}_{b} \right) \quad (48)$$

where $n_{L_{k-1}}^{FoV}$ refers to the number of FoV landmarks. (48) can be bounded by expanding the failure event space; ignoring b :

$$P(MA_i) \leq P\left(\bigcup_{j=1, j \neq i}^{n_{L_{k-1}}^{FoV}} \left\| \mathbf{y}_{j,i} \right\|_{\mathbf{Y}_j^{-1}} < T_a\right) \quad (49)$$

By replacing the probability of the union of events with the summation of the individual probabilities of each event, (48) can be upper-bounded as follows (Arana et al., 2020a), that is:

$$P(MA_i) \leq \sum_{j=1, j \neq i}^{n_{L_{k-1}}^{FoV}} P\left(\left\| \mathbf{y}_{j,i} \right\|_{\mathbf{Y}_j^{-1}} < T_a\right) \quad (50)$$

By using the definition of the individual innovation in (3) and linearizing the measurement's function, (50) is expanded as follows:

$$\begin{aligned} P(MA_i) &\leq \sum_{j=1, j \neq i}^{n_{L_{k-1}}^{FoV}} P\left(\left\| \mathbf{z}_j - \mathbf{h}_{l_i}(\bar{\mathbf{x}}^a) \right\|_{\mathbf{Y}_j^{-1}} < T_a\right) \\ &\leq \sum_{j=1, j \neq i}^{n_{L_{k-1}}^{FoV}} P\left(\left\| \mathbf{h}_{l_j}(\bar{\mathbf{x}}^a) + \mathbf{v}_j - \mathbf{h}_{l_i}(\bar{\mathbf{x}}^a) \right\|_{\mathbf{Y}_j^{-1}} < T_a\right) \\ &\leq \sum_{j=1, j \neq i}^{n_{L_{k-1}}^{FoV}} P\left(\left\| \mathbf{H}_{l_j} \delta \bar{\mathbf{x}}^a + \mathbf{v}_j + \mathbf{y}_{j,i} \right\|_{\mathbf{Y}_j^{-1}} < T_a\right) \end{aligned} \quad (51)$$

where $\mathbf{y}_{j,i} = \mathbf{h}_{l_j}(\bar{\mathbf{x}}^a) - \mathbf{h}_{l_i}(\bar{\mathbf{x}}^a)$ is the landmark separation metric. (51) is upper-bounded by utilizing the trigonometric inequality $\|\mathbf{a} + \mathbf{b}\| \geq \|\mathbf{a}\| - \|\mathbf{b}\|$ as follows (Arana et al., 2020a), that is:

$$P(MA_i) \leq \sum_{j=1, j \neq i}^{n_{L_{k-1}}^{FoV}} P\left(\left\| \mathbf{y}_{j,i} \right\|_{\mathbf{Y}_j^{-1}} - T_a < \left\| \mathbf{H}_{l_j} \delta \bar{\mathbf{x}}^a + \mathbf{v}_j \right\|_{\mathbf{Y}_j^{-1}}\right) \quad (52)$$

By squaring the inequality's sides, $P(MA_i)$ bound turns to:

$$P(MA_i) \leq \sum_{j=1, j \neq i}^{n_{L_{k-1}}^{FoV}} P\left(\left(\left\| \mathbf{y}_{j,i} \right\|_{\mathbf{Y}_j^{-1}} - T_a\right)^2 < \left\| \mathbf{H}_{l_j} \delta \bar{\mathbf{x}}^a + \mathbf{v}_j \right\|_{\mathbf{Y}_j^{-1}}^2\right) \quad (53)$$

Recall from (3) that $\mathbf{Y}^j$ is the covariance of $\mathbf{H}_{l_j} \delta \bar{\mathbf{x}}^a + \mathbf{v}_j$. Thus, the distribution of the inequality's right side in (53) is:

$$\left\| \mathbf{H}_{l_j} \delta \bar{\mathbf{x}}^a + \mathbf{v}_j \right\|_{\mathbf{Y}_j^{-1}}^2 \sim \chi_{m_F, \mu_j}^2 \quad (54)$$

where μ_j is the non-centrality parameter, defined as follows:

$$\begin{aligned} \mu_j &= \left\| \mathbf{H}_{l_j} \bar{\mathbf{f}}^a \right\|_{\mathbf{Y}_j^{-1}}^2 = \left\| \mathbf{H}_{l_j} \mathbf{G}_{k-1}^a \hat{\mathbf{f}}_{k-1}^a \right\|_{\mathbf{Y}_j^{-1}}^2 \\ &= \left\| \check{\mathbf{f}}_{k-1}^a \right\|_{\underbrace{\mathbf{A}_{k-1}^T \mathbf{G}_{k-1}^a \mathbf{H}_{l_j}^T \mathbf{Y}_j^{-1} \mathbf{H}_{l_j} \mathbf{G}_{k-1}^a \mathbf{A}_{k-1}}_{=\mathbf{Q}^j}}^2 \end{aligned} \quad (55)$$

Note that μ_j is unknown because the combined fault vector at the previous time epoch, $\check{\mathbf{f}}_{k-1}$, is unknown. As will be

shown in the following section, μ_j will be upper-bounded for each sensor measurement's fault hypothesis of the previous time epoch. Accordingly, the $P(MA_i)$ bound, in (53), can be expanded by utilizing the total probability theorem as follows:

$$P(MA_i) \leq \sum_{j=1, j \neq i_l}^{n_{FoV}^{L_{k-1}}} \sum_{h=0}^{n_H} P(H_{k-1}^h) P\left(\left(\|\mathbf{y}_{j,i}\|_{\mathbf{y}_j^{-1}} - T_a\right)^2 < \|\mathbf{H}_j \delta \bar{\mathbf{x}}^a + \mathbf{v}_j\|_{\mathbf{y}_j^{-1}}^2 \middle| H_{k-1}^h\right) \quad (56)$$

Given (54) and (55) for each fault hypothesis of the previous time epoch, $P(MA_i)$ bound, in (56), can be expressed as follows, that is:

$$P(MA_i) \leq n_{L_{k-1}}^{FoV} - 1 - \sum_{j=1, j \neq i_l}^{n_{L_{k-1}}^{FoV}} \sum_{h=0}^{n_H} X_{mF, \mu_j, h}^2 \left[\left(\|\mathbf{y}_{j,i}\|_{\mathbf{y}_j^{-1}} - T_a \right)^2 \right] P(H_{k-1}^h) \quad (57)$$

Note that the only unknown is $\mu_{j,h}$ as it is a function of the combined fault vector at the former epoch; see (55). Thus, an upper-bound on $\mu_{j,h}$ will be derived in the following section.

Because this subsection has been focused on quantifying the risk of failure for each feature from a mathematical perspective, an intuitive understanding of the impact of both feature data association errors on localization performance will be demonstrated. Note, from (25), that the state estimate error is a function of the innovation. The innovation is the random variable might be affected by faults. From (51), these faults become non-zero when the mis-association and the unmapped-association events occur. Specifically, this occurs when $\mathbf{y}_{j,i} \neq \mathbf{0}$, that is, the i^{th} extracted feature is incorrectly associated to the j^{th} landmark; if the i^{th} extracted feature is extracted from a landmark in the map, the mis-association occurs, whereas if the i^{th} extracted feature is extracted from an unmapped object, the unmapped-association takes place.

7.2. Landmark's failure probability

Since landmarks are estimated by the sequential association of the extracted feature measurements and that incorrectly associating features to a landmark leads to a faulted estimate of that landmark, the landmark failure risk will be quantified sequentially whenever a new feature is associated to it. The modes of failure in adding new landmarks are as follows, that is:

1. *Mis-addition* (Ma_j) happens when a feature measurement that corresponds to a pre-estimated landmark in the map is utilized to add a new landmark in the map.
2. *Dynamic-addition* (Da_j) happens when a feature measurement that corresponds to a unmapped moving object is utilized to add a new landmark in the map.

To simplify notation, the landmark's probability of failure, $P(F_{k-1}^{l_j})$, is derived by assuming that the landmark is added at the first time epoch and is associated at all time epochs as follows:

$$P(F_{k-1}^{l_j}) = P\left(\underbrace{Ma_j \cup Da_j}_a \bigcup_{r=1}^{k-1} \underbrace{[UA_{j_l, r} \cup MA_{j_l, r}]}_{b_r}\right) \quad (58)$$

where j_l refers to the feature's measurement associated to the j^{th} estimated landmark. Note that a represents the event of mistakenly adding the j^{th} estimated landmark, whereas b_r represents the event of mistakenly associating features to it at time epoch r . By utilizing the probabilistic inequality, $P(A \cup B) \leq P(A) + P(B)$, $P(F_{k-1}^{l_j})$ can be upper-bounded as follows:

$$\begin{aligned} P(F_{k-1}^{l_j}) &= P\left(\underbrace{Ma_j \cup Da_j}_a \bigcup_{r=1}^{k-1} \underbrace{[UA_{j_l, r} \cup MA_{j_l, r}]}_{b_r}\right) \\ &= P\left(Ma_j \cup Da_j \bigcup_{r=1}^{k-2} [UA_{j_l, r} \cup MA_{j_l, r}] \cup UA_{j_l, k-1} \cup MA_{j_l, k-1}\right) \\ &\leq P\left(Ma_j \cup Da_j \bigcup_{r=1}^{k-2} [UA_{j_l, r} \cup MA_{j_l, r}]\right) + P(UA_{j_l, k-1} \cup MA_{j_l, k-1}) \\ &\leq P(F_{k-2}^{l_j}) + P(F_k^{j_l}), \text{ where } P(F_0^{l_j}) = P(Ma_j \cup Da_j) \end{aligned} \quad (59)$$

where the failure risk of the feature associated with the j^{th} estimated landmark is upper-bounded in Sections 7-A. Note that the failure risk of the j^{th} estimated landmark is quantified recursively. The only remaining unknown parameter, in (59), is the risk of incorrectly adding the j^{th} estimated landmark's.

$P(F_0^{l_j})$, which refers to the probability of the incorrect addition of the j^{th} estimated landmark, is upper-bounded by replacing the probability of the events' union of Ma_j and Da_j by the summation of their individual probabilities as follows:

$$P(F_0^{l_j}) \leq P(Ma_j) + P(Da_j) \quad (60)$$

This paper considers $P(Da_j)$ a constant that can be upper-bounded experimentally. Recall that Ma_j occurs if the j^{th} non-associated feature measurement that corresponds to a pre-estimated landmark is utilized to add a new estimated landmark. From (5), $P(Ma_j)$ can be written as follows, that is:

$$P(Ma_j) = P\left(\bigcap_{l=1}^{n_{FoV}^L} \|\mathbf{y}_{j,l,0}\|_{\mathbf{y}_{j,0}^{-1}} > T_n\right) \quad (61)$$

Because the landmark, from which the j^{th} non-associated feature's measurement (utilized to add a new landmark) is extracted, is unknown, j^* , $P(Ma_j)$ can be upper-bounded by using the estimated landmarks in the robot's FoV as follows:

$$P(Ma_j) \leq P\left(\bigcup_{r=1}^{n_L^{FoV}} \left(\bigcap_{l=1}^{n_L^{FoV}} \|\mathbf{y}_{r,l,0}\|_{\mathbf{y}_{r,0}^{-1}} > T_n\right)\right) \quad (62)$$

By replacing the probability of the events' union with the sum of their individual probabilities, $P(Ma_j)$ is bounded as follows:

$$P(Ma_j) \leq \sum_{r=1}^{n_L^{FoV}} P\left(\bigcap_{l=1}^{n_L^{FoV}} \|\mathbf{y}_{r,l,0}\|_{\mathbf{y}_{r,0}^{-1}} > T_n\right) \quad (63)$$

(63) is upper-bounded by ignoring the events in which $l \neq l_r$:

$$P(Ma_j) \leq \sum_{r=1}^{n_L^{FoV}} P\left(\|\mathbf{y}_{r,l_r,0}\|_{\mathbf{y}_{r,0}^{-1}} > T_n\right) \quad (64)$$

Algorithm 1 EKF based-SLAM integrity monitoring

- 1: Given:
 - 2: Prior state estimate and covariance, $\hat{\mathbf{x}}_{k-1}$ and $\hat{\mathbf{P}}_{k-1}$.
 - 3: Current control-input with its covariance, $\mathbf{u}_{k-1}$ and $\mathbf{W}_k$.
 - 4: Current measurement and its covariance, $\mathbf{z}_k$ and $\mathbf{V}$.
 - 5: The models $\mathbf{g}(\cdot, \cdot)$, $\mathbf{h}(\cdot, \cdot)$ and $\Phi(\cdot, \cdot)$ in (1), (2) and (16).
 - 6: Predict state estimate and its covariance, in (6) and (7).
 - 7: Associate n_a^F features to estimated landmarks using (4).
 - 8: Update the estimate and the covariance, in (11) and (12).
 - 9: Evaluate the detector and its threshold via (21) and (29).
 - 10: Utilize n_a^F features to add landmarks using (18) and (19).
 - 11: **if** the alarm is not triggered **then**
 - 12: **for** Each landmark, j , in FoV and hypothesis, h **do**
 - 13: Upper-bound on $\mu_{j,h}$ using (70), (75) and (69).
 - 14: Evaluate an upper-bound on $P(Ma_j)$ using (77).
 - 15: **for** Each associated feature, i **do**
 - 16: Evaluate an upper-bound on $P(MA_i)$ using (76).
 - 17: Evaluate an upper-bound on $P(F^{z_i})$ using (47)
 - 18: **for** Each estimated landmark, l_j **do**
 - 19: Evaluate $P(F_{k-1}^{l_j})$ using (59).
 - 20: **for** Fault hypothesis, h **do**
 - 21: Evaluate a bound on $P(H^h)$ using (42) and (43).
 - 22: Evaluate $P(HMI|H^h)$ bound using (38) and (39).
 - 23: Quantify a bound on $P(HMI)$ using (44) and (45).
-

Using the definition of the individual innovation in (3) and linearizing the measurement function, (64) can be written as follows:

$$\begin{aligned} P(Ma_j) &\leq \sum_{r=1}^{n_L^{FoV}} P\left(\|\mathbf{z}_{r,0} - \mathbf{h}_{l_r}(\bar{\mathbf{x}}_0^a)\|_{\mathbf{y}_{r,0}^{-1}} > T_n\right) \\ &\leq \sum_{r=1}^{n_L^{FoV}} P\left(\|\mathbf{h}_{l_r}(\mathbf{x}_0^a) + \mathbf{v}_{r,0} - \mathbf{h}_{l_r}(\bar{\mathbf{x}}_0^a)\|_{\mathbf{y}_{r,0}^{-1}} > T_n\right) \\ &\leq \sum_{r=1}^{n_L^{FoV}} P\left(\|\mathbf{H}_0^l \delta \bar{\mathbf{x}}_0^a + \mathbf{v}_{r,0}\|_{\mathbf{y}_{r,0}^{-1}} > T_n\right) \end{aligned} \quad (65)$$

Squaring the inequality's sides, $P(Ma_j)$ bound reduces to:

$$P(Ma_j) \leq \sum_{r=1}^{n_L^{FoV}} P\left(\underbrace{\|\mathbf{H}_0^l \delta \bar{\mathbf{x}}_0^a + \mathbf{v}_{r,0}\|_{\mathbf{y}_{r,0}^{-1}}^2}_a > T_n^2\right) \quad (66)$$

Recall that the distribution of the term a is a function of the combined fault at the previous epoch, $\check{\mathbf{f}}_{k-1}$; see (54) and (55). Accordingly, the total probability rule is utilized to expand $P(Ma_j)$ over the hypotheses at the former epoch as follows:

$$P(Ma_j) \leq \sum_{r=1}^{n_L^{FoV}} \sum_{h=0}^{n_H} P\left(\|\mathbf{H}_0^l \delta \bar{\mathbf{x}}_0^a + \mathbf{v}_{r,0}\|_{\mathbf{y}_{r,0}^{-1}}^2 > T_n^2 \mid H_{k-1}^h\right) P(H_{k-1}^h) \quad (67)$$

Given (54) and (55) for each fault hypothesis of the former time epoch, $P(Ma_j)$, in (67), can be written as follows, that is:

$$P(Ma_j) \leq n_L^{FoV} - \sum_{r=1}^{n_L^{FoV}} \sum_{h=0}^{n_H} X_{m_F, \mu_{r,h}}^2 [T_n^2] P(H_{k-1}^h) \quad (68)$$

Note that the only unknown is $\mu_{j,h}$ as it is a function of the previous combined fault, see (55), that will be bounded next.

8. Bounding non-centrally parameter

The non-centrality $\mu_{j,h}$ represents the impact of prior faults on the risk of feature and landmark failure. Because sensor faults are unknown, an upper-bound on $\mu_{j,h}$ will be derived. This is done by using the fact that the detector did not trigger an alarm at the prior epoch, $q_{k-1} \leq T_{k-1}$, as the two detector non-centralities at the prior epoch, λ_{k-1} , and $\mu_{j,h}$ are affected by the combined fault vector, $\check{\mathbf{f}}_{k-1}$; see (28) and (55). Via $\Gamma_{j,h}$ as a linear scaling factor, λ_{k-1} and $\mu_{j,h}$ are related as follows:

$$\mu_{j,h} = \Gamma_{j,h} \lambda_{k-1,h} \quad (69)$$

Arana et al. (2018) has derived an upper-bound for the non-centrality parameter of any chi-squared-based fault detection mechanism by using confidence intervals, such that the failure probability for this bound is less than a predefined risk, I_{NC} . Accordingly, given that the non-detection event, $q_{k-1} \leq T_{k-1}$, an upper-bound on $\lambda_{k-1,h}$ can be expressed as follows:

$$\check{\lambda}_{k-1,h} = \left(\sqrt{T_{k-1}} + \sqrt{X_{n_{k-1}}^{-2} [1 - I_{NC}]}\right)^2 \geq \lambda_{k-1} \quad (70)$$

I_{NC} will be accounted for in $P(MA_k^i)$ and $P(Ma_j)$ later. By substituting (70) in (69), the bound on $\mu_{j,h}$ is written as follows:

$$\check{\mu}_h^j = \check{\Gamma}_h^j \check{\lambda}_{k-1} \geq \mu_{j,h} \quad (71)$$

where $\check{\Gamma}_h^j$ is the upper-bound on the scaling factor, $\Gamma_{j,h}$, that is to be derived next. Via (28) and (55), $\Gamma_{j,h}$ is expressed as follows:

$$\Gamma_{j,h} = \frac{\mu_{j,h}}{\lambda_{k-1,h}} = \frac{\check{\mathbf{f}}_{k-1}^T \mathbf{A}_{k-1}^T \mathbf{G}_{k-1}^{aT} \mathbf{H}_{lj}^T \mathbf{Y}_j^{-1} \mathbf{H}_{lj} \mathbf{G}_{k-1}^a \mathbf{A}_{k-1} \check{\mathbf{f}}_{k-1}}{\check{\mathbf{f}}_{k-1}^T \mathbf{B}_{k-1}^T \mathbf{Y}_{k-1}^{-1} \mathbf{B}_{k-1} \check{\mathbf{f}}_{k-1}} \quad (72)$$

Because only the faulted measurements in the hypothesis H_{k-1}^h affect $\mu_{j,h}$ and $\lambda_{k-1,h}$, $\Gamma_{j,h}$ can be expanded as follows:

$$\begin{aligned} \Gamma_{j,h} &= \frac{\check{\mathbf{f}}_{k-1}^T \mathbf{E}_{k-1,h}^T \mathbf{E}_{k-1,h} \mathbf{A}_{k-1}^T \mathbf{G}_{k-1}^{aT} \mathbf{H}_{lj}^T \mathbf{Y}_j^{-1} \mathbf{H}_{lj} \mathbf{G}_{k-1}^a \mathbf{A}_{k-1} \mathbf{E}_{k-1,h} \check{\mathbf{f}}_{k-1}}{\check{\mathbf{f}}_{k-1}^T \mathbf{E}_{k-1,h}^T \mathbf{E}_{k-1,h} \mathbf{B}_{k-1}^T \mathbf{Y}_{k-1}^{-1} \mathbf{B}_{k-1} \mathbf{E}_{k-1,h} \check{\mathbf{f}}_{k-1}} \\ &= \frac{\check{\mathbf{f}}_{k-1}^T \mathbf{E}_{k-1,h}^T \mathbf{N}_h^{1/2} \mathbf{N}_h^{-1/2} \mathbf{M}_h^j \mathbf{N}_h^{-1/2} \mathbf{N}_h^{1/2} \mathbf{E}_{k-1,h} \check{\mathbf{f}}_{k-1}}{\check{\mathbf{f}}_{k-1}^T \mathbf{E}_{k-1,h}^T \mathbf{N}_h \mathbf{E}_{k-1,h} \check{\mathbf{f}}_{k-1}} \quad (73) \end{aligned}$$

$\Gamma_{j,h}$ is bounded via the fact that the subtraction of a positive semi-definite matrix from a diagonal matrix with its elements as the spectral radius, $\rho(\cdot)$, is positive semi-definite as, that is:

$$\rho(\mathbf{N}_h^{-1/2} \mathbf{M}_h^j \mathbf{N}_h^{-1/2}) \mathbf{I} - \mathbf{N}_h^{-1/2} \mathbf{M}_h^j \mathbf{N}_h^{-1/2} \geq \mathbf{0} \quad (74)$$

Thus, the upper bound on $\Gamma_{j,h}$ can be expressed as follows:

$$\begin{aligned} \check{\Gamma}_h^j &= \frac{\check{\mathbf{f}}_{k-1}^T \mathbf{E}_{k-1,h}^T \mathbf{N}_h^{1/2} \rho(\mathbf{N}_h^{-1/2} \mathbf{M}_h^j \mathbf{N}_h^{-1/2}) \mathbf{N}_h^{1/2} \mathbf{E}_{k-1,h} \check{\mathbf{f}}_{k-1}}{\check{\mathbf{f}}_{k-1}^T \mathbf{E}_{k-1,h}^T \mathbf{N}_h \mathbf{E}_{k-1,h} \check{\mathbf{f}}_{k-1}} \\ &= \rho(\mathbf{N}_h^{-1/2} \mathbf{M}_h^j \mathbf{N}_h^{-1/2}) \frac{\check{\mathbf{f}}_{k-1}^T \mathbf{E}_{k-1,h}^T \mathbf{N}_h \mathbf{E}_{k-1,h} \check{\mathbf{f}}_{k-1}}{\check{\mathbf{f}}_{k-1}^T \mathbf{E}_{k-1,h}^T \mathbf{N}_h \mathbf{E}_{k-1,h} \check{\mathbf{f}}_{k-1}} \\ &= \rho(\mathbf{N}_h^{-1/2} \mathbf{M}_h^j \mathbf{N}_h^{-1/2}) \quad (75) \end{aligned}$$

Now, $\check{\mu}_h^j$, in (71), will be substituted back in (57) and (68):

$$\begin{aligned} P(MA_i) &\leq I_{NC} + \\ n_{L_{k-1}}^{FoV} &- 1 - \sum_{j=1, j \neq i_l}^{n_{L_{k-1}}^{FoV}} \sum_{h=0}^{n_H} X_{m_F, \check{\mu}_h^j}^2 \left[\left(\|\mathbf{y}_{j,i}\|_{\mathbf{Y}_j^{-1}} - T_a \right)^2 \right] P(H_{k-1}^h) \quad (76) \end{aligned}$$

$$P(Ma_j) \leq I_{NC} + n_L^{FoV} - \sum_{r=1}^{n_L^{FoV}} \sum_{h=0}^{n_H} X_{m_F, \check{\mu}_h^r}^2 [T_n^2] P(H_{k-1}^h) \quad (77)$$

where I_{NC} is added to account for the risk of failure for the non-centrality's upper bound of the detector, $\lambda_{k-1,h}$, in (70).

Algorithm 1 summarizes the main steps needed to monitor integrity. Next, simulation and experimentation results will be utilized to assess the proposed localization safety metric.

9. Results

This section investigates the performance of the proposed safety evaluation method for EKF-SLAM while considering incorrect data association errors in features, mis-association and unmapped-association, and landmarks, mis-addition and dynamic-addition, using both simulations and experiments.

9.1. Simulation

A robot is simulated via a constant-speed kinematic bicycle model traversing predefined way-points' set in a landmark-rich environment, see Figure 2. The rear wheel's velocity

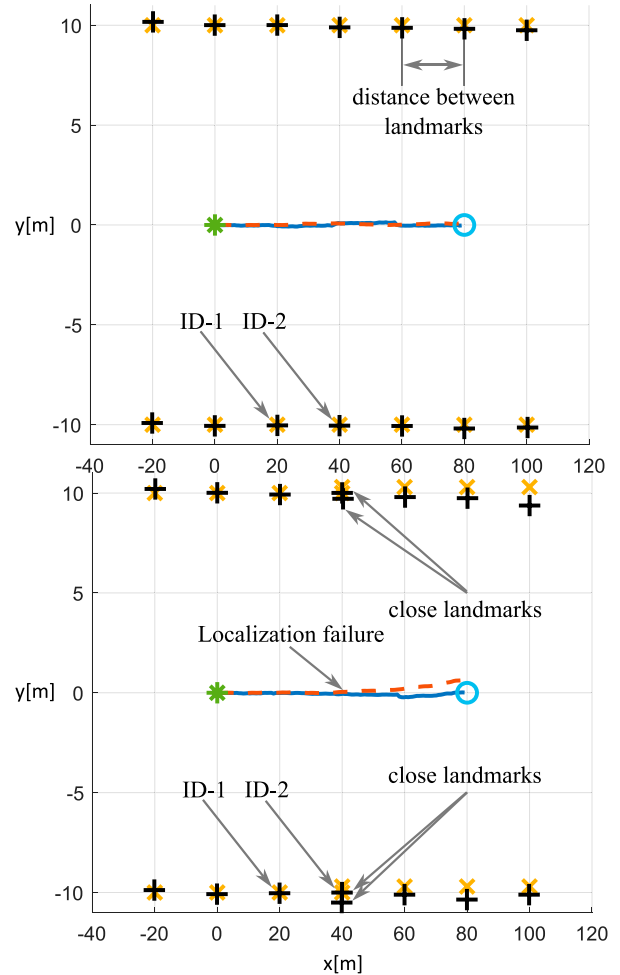


Figure 2. Red and blue lines are estimated and true trajectories while traversing two maps, respectively. Black pluses and yellow crosses are estimated and true landmarks, respectively. Green asterisk and cyan circle are the robot starting and ending points.

and steering angle represent the relative measurements, whereas the mobile robot range and bearing to landmarks are absolute measurements. Table 1 displays the simulation's parameters.

The results of the simulation have been decomposed into two portions. The first part investigates the supremacy of the derived localization integrity risk metric over the commonly used 2σ covariance envelope in accounting for faults. The second part studies the relationship between the density of landmarks in the environment and robot localization safety.

To illustrate the importance of the proposed localization safety quantification method over the traditional covariance bound, two simulation's environments have been configured such that all landmarks are properly-separated in the first one (see Figure 2; top), whereas the second environment contains two close landmarks at $x = 40$ m, on each side of the track (see Figure 2; bottom). Note that, in the second environment, the landmarks, that follow the close landmarks, have been shifted such that any data association errors happening at $x = 40$ m will not be reflected back on the innovation vector at later time epochs, and so the robot localization error, in the state-of-interest, will increase without being reflected on the 2σ covariance envelope, see Figure 3. Moreover, note that the 2σ covariance envelope tends to be tighter when there are more feature measurements; however, one caveat is that it does not consider sensor measurement faults. Since the large lateral positioning error has not been reflected on the innovation vector in the second environment, the fault alarm failed to detect the data association error, see Figure 4. Note that the failure detector's threshold tend to be higher when there are more features, as the degrees of freedom for the detector is higher. Figure 5 shows the localization integrity risk for each

landmark map. Note, in the second environment, the integrity risk peaks at $x = 20$ m when the two close landmarks enter the mobile robot field of view. Thus, unlike the 2σ covariance envelope bound, the proposed integrity risk metric succeeds in predicting mobile robot localization failure. However, to visually understand the reason behind the outstanding performance of the integrity risk mechanism, in predicting localization failure, the failure probability of the estimated landmarks, and their extracted features, that is labeled with ID-1 and ID-2, in each landmark map, is displayed in Figure 6. Note that when the robot approaches $x = 20$ m, that is, the estimated landmark ID – 2 becomes within the sensor FoV, the failure probability of the landmark ID – 2 jumps up to one in the second environment due its improper placement with respect to the surrounding landmarks, unlike the landmark ID – 1, making the robot's localization integrity risk to approach one. This behavior matches the trend of the true state-of-interest estimate error.

To investigate the relationship between the environment's landmark density in the map and localization safety, a robot is simulated traversing 15 different landmark maps. Each landmark map is similar to the one in Figure 2 (top) except that the distance between landmarks is varied from 8 m to 23 m in 1 m increments such that the number of landmarks in the map increases when the distance between landmarks decreases.

Localization's availability is defined as the percentage of time in which the integrity risk is below a safety requirement, selected here to be $1e - 5$ (Reid et al., 2019). Figure 7 shows the robot's localization availability for each of the 15 simulated runs. Note, the localization availability increases as the distance between map's landmarks decreases, that is, the landmark density increases, as this reflects more

Table 1. Simulation parameters.

Speed	25 km/h	$\sigma_{velocity}$	0.1 m/s
Time step	0.1 s	$\sigma_{steer\ angle}$	0.4°
Sensor range	25 m	σ_{lidar}	0.2 m
Alert limit	1.5 m	I_{FA}	10^{-5}
P_{UA}, P_{Da}	10^{-15}	Wheelbase	1.0 m

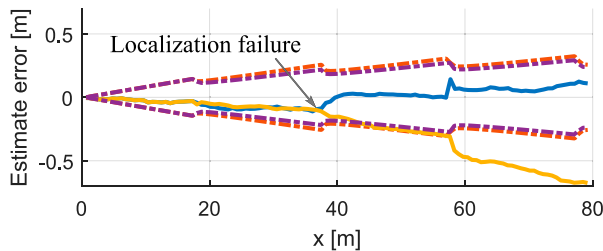


Figure 3. Lateral robot estimate error (blue) with its 3σ bounds (dashed red) for the properly-spaced landmark map, and lateral robot estimate error (yellow) with its 3σ bounds (purple) for the poorly-spaced map versus the traveled distance in X -direction.

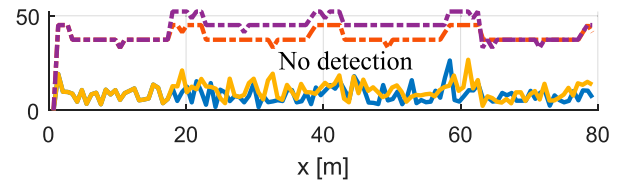


Figure 4. The detector (blue) along with its threshold (dashed red) for the nominally-spaced landmark map, and the detector (yellow) along with its threshold (dashed purple) versus the true traveled distance in X -axis for the poorly-spaced landmark map.

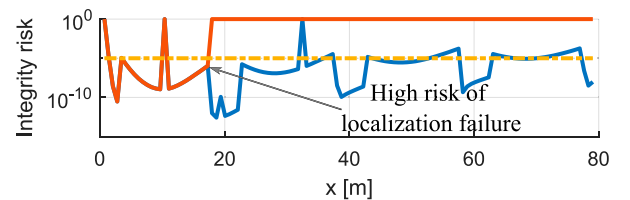


Figure 5. The integrity risk for the nominally-spaced (upper) landmark map (blue) and for the poorly-spaced (lower) landmark map (red), in addition to the requirement (dashed yellow) versus the true robot's traveled distance in X -direction.

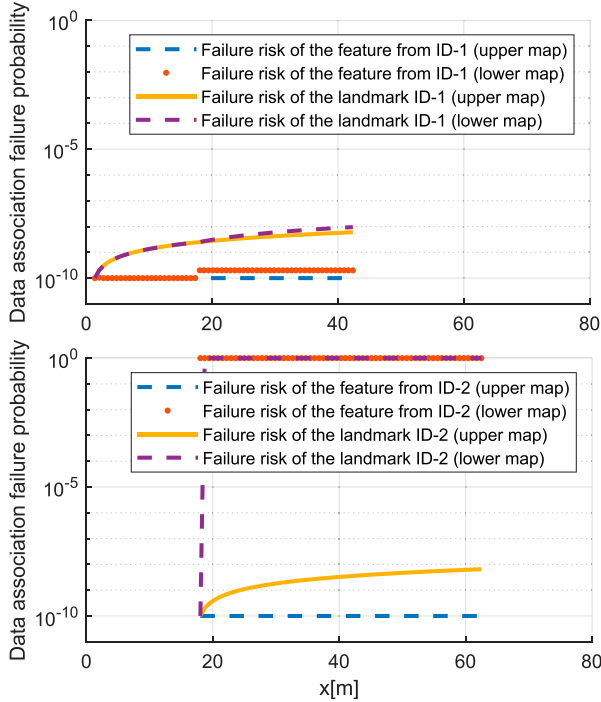


Figure 6. The failure probability of the estimated landmarks, that is labeled with ID-1 and ID-2, and their extracted features for the two maps versus the true robot's position in X -direction.

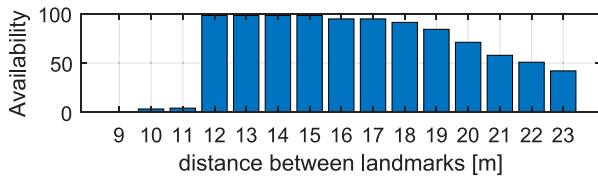


Figure 7. Localization availability versus the longitudinal distance between every two consecutive landmarks in the map.

information about the robot state. However, if the map's landmark density becomes relatively high, then the localization's availability drastically drops because the risk of data association errors dominates over the increase in the number of measurements.

9.2. Experimentation

The performance of the introduced robot localization safety method for EKF-SLAM has been examined in an urban area, S. Federal Street from 31st to 33rd St., in Chicago, IL USA where GPS coverage is degraded or denied because of the existence of dense tree clusters along the street (Nagai et al., 2024); see Figure 8. The sensor suite mounted on a vehicle's roof rack (see Figure 9) is comprised of a STIM-300 tactical inertial measurement's unit for relative measurements, and Ouster lidar for range and bearing measurements to columns and tree trunks that represent the absolute measurements and the set of landmarks to be



Figure 8. The transect utilized consists of S. Federal street from 31st to 33rd within Illinois tech campus in Chicago, IL.



Figure 9. The testing setup utilized in the experiment. The sensor suite comprises two velodyne VLP-16 lidars, one ouster-64 lidar, STIM-300 tactical-grade IMU, and novatel SPAN-CPT DGPS attached to a roof rack of a vehicle. In this experiment, only the ouster lidar and STIM-300 were utilized. Lamp posts, columns, and tree trunks are used as landmarks.

mapped. In this experiment, GPS receiver measurements have not been utilized because even when there is coverage, multi-pathing errors are prevalent.

The vehicle's state vector contains 15 variables: three for the location, three for the attitude, three for the linear velocities, and six for the gyroscopes' and accelerometers' biases. Table 2 shows the used experiment's parameters. To investigate the ability of the proposed method in predicting mobile robot's localization failure, the test scenario has been selected such that EKF-based SLAM would fail during the experiment, due to data association errors. As a ground truth, NDT scan matching-based batch estimation algorithm was utilized Jiang et al. (2023). Figure 10 shows the ground truth's and the estimated vehicle's trajectory and map of landmarks.

To reduce the impact of faulted landmarks on the mobile robot's state estimate, the landmarks with failure risk more than 1% is not associated in the local nearest neighbor data association algorithm. This is a useful isolation tool for the landmarks with high failure risk, which will eliminate the impact of their data association error on the positioning performance of EKF-SLAM, and so improve the localization

Table 2. Experiment parameters.

PSD _{AccNoise}	0.002 m ² /s ⁵	Alert limit	1.5 m
PSD _{GyroNoise}	3.1e − 06 rad ² /s ³	σ_{lidar}	0.25 m
PSD _{AccBias}	2.4e − 07 m ² /s ⁶	freq _{lidar}	10 Hz
PSD _{GyroBias}	2e − 12 rad ² /s ⁴	freq _{IMU}	125 Hz
I_{FA}	10 ^{−5}	P_{UA}	10 ^{−10}

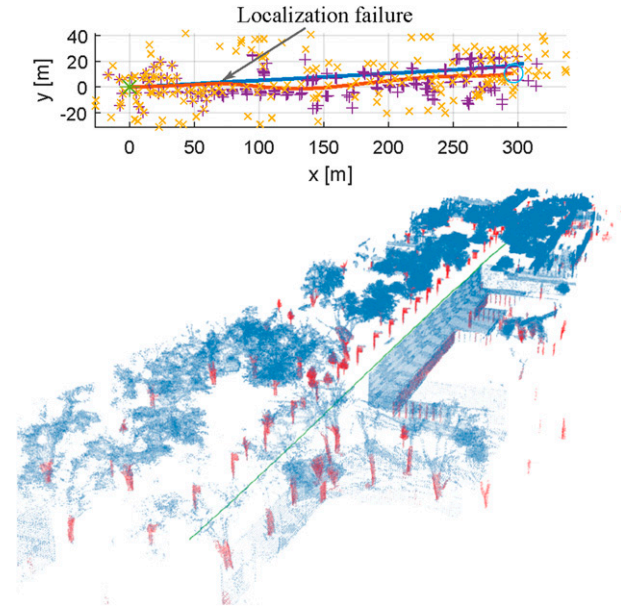


Figure 10. (Top figure) The vehicle's estimated trajectory using EKF-SLAM red line. The ground-truth trajectory that was obtained via NDT scan matching-based batch estimation (blue line). The purple plus signs and the yellow cross signs represent the mapped landmarks (lamp posts, columns, and tree trunks) via EKF-SLAM and NDT batch estimation, respectively. The green asterisk refers to the vehicle's starting location, whereas the cyan circle is the ending location. (Bottom Figure) The ground-truth vehicle's trajectory (green) in a 3D point cloud environment (blue) of landmarks to be estimated (red).

integrity risk. Simply, to utilize this tool, the failure events of these landmarks to be isolated and their associated features will be excluded from the fault hypotheses, in addition, the features associated to these landmarks will not be included in the innovation vector in EKF-based SLAM's update step.

In this paper, the state of interest is designated to be the vehicle lateral offset. From Figure 10, note that EKF-based SLAM failure takes place when the vehicle becomes at 60 m in the x -direction, that is, the difference between the vehicle's estimate and the ground truth becomes large. The reason for this localization failure is the combination of two events: (1) the lack of sufficient redundancy (few number of extracted features) and (2) the close proximity between the estimated landmarks in the map by means of the Mahalanobis distance.

Figure 11 shows the localization's integrity risk along with the safety threshold for the experimental scenario. Note that the localization integrity risk becomes above the

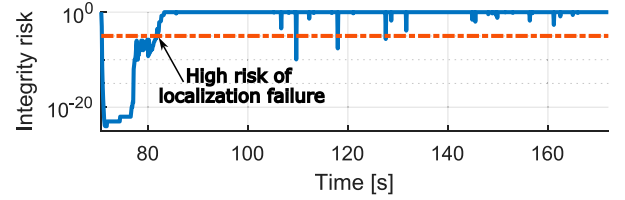


Figure 11. The localization integrity risk (blue line) and the predefined safety requirement (dashed red) versus time.

predefined safety threshold after the event of robot localization failure, reflecting that the proposed localization safety quantification method is able to successfully predict localization's failure.

10. Conclusions

This paper introduces a mobile robot's localization safety quantification method for EKF-based SLAM state estimation algorithm. The results show that the technique is capable of predicting the impact of data association errors on mobile robot's localization performance. In addition, the proposed positioning safety evaluation method suggests that mobile robot's localization performance improves as the landmark density in the map increases as long as the landmarks are distinguishable by the local nearest neighbors criterion.

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